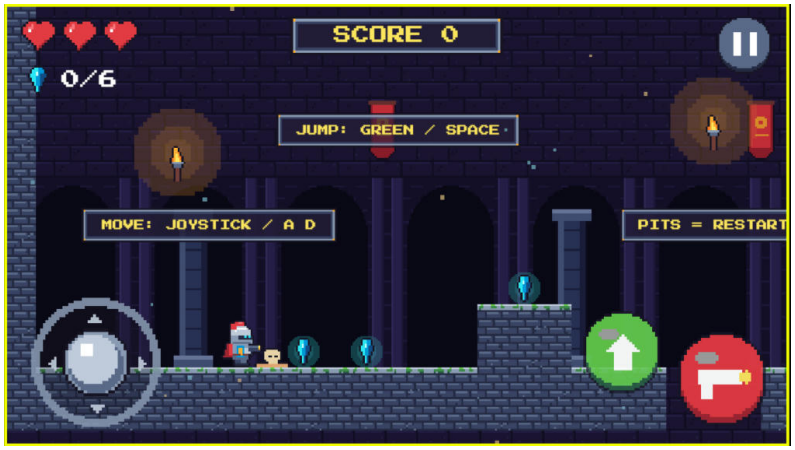
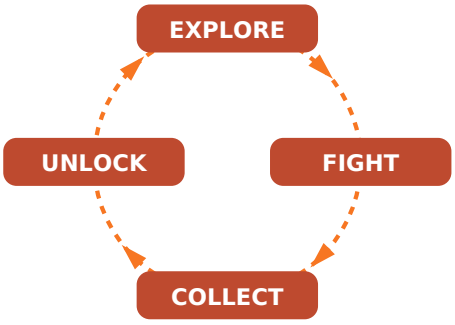
Dungeon Knight

GDD v1 · Game Development Fundamentals
Weeks 9–13 · solo project (teams form in week 14)



Level 1 HUD (Device Simulator, 16:9): hearts and crystals top-left, score top-center, pause top-right, joystick bottom-left, JUMP and FIRE bottom-right.



Core loop — repeats every 30–60 s

1 · TITLE + PITCH	**Dungeon Knight: Rise of the Slime King.** A tiny knight with a blaster runs and jumps through a pixel dungeon, shoots slimes and bats, collects energy crystals to open portals, and defeats the Slime King.
2 · GENRE · PLATFORM	Action platformer with shooting (side view). **iOS and Android**, phone and tablet, **landscape**. Tested on iPhone (Unity Remote 5) and in Device Simulator: 16:9, 19.5:9 notch phone, 4:3 tablet.
3 · 2D OR 3D	**2D.** Sprite hero with Idle / Run / Jump / Fall animations and flipX. Tilemap levels, Rigidbody2D physics, side camera.
4 · CORE LOOP	**Explore** the level → **Fight** enemies → **Collect** crystals → **Unlock** the portal to a harder level. Every few levels a boss tests everything the player learned.
5 · CONTROLS	Bottom-left: **on-screen joystick** — run left / right. Bottom-right: **JUMP** (green, hold = higher jump) and **FIRE** (red, hold = auto-fire with cooldown). Top-right: **PAUSE** (resume · restart · menu). Keyboard for debugging: A/D, Space, X, Esc. Unity Input System, On-Screen Stick / Button.
6 · CONTENT	MVP build: **5 levels** in 2 worlds (Crypt, Deep Halls) + **1 boss** arena. **4 enemy types**: patrol slime, flying bat, chasing slime (week 10 AI), goblin archer (week 10 AI). **Items**: energy crystal (score + portal key), heart (heal). **Hazards**: spikes, pits, falling stalactites, moving platforms.
7 · ART AND SOUND	Soul Knight-inspired 16 px pixel art, dark dungeon palette with bright neon shots. All sprites are original (drawn procedurally for this project), font Press Start 2P (OFL). Original chiptune music for menu / levels / boss and 15+ sound effects. One world: knight, slimes, crystals, UI frames all share the same palette and pixel size.
8 · MVP (WEEK 15)	Menu → level select → 5 levels → boss → win. Pause menu. HUD that fits every phone (scaler, anchors, safe area). Enemy AI with states. **Saving**: unlocked levels, best score, volume. Settings (music / SFX). Sound mixer, particle VFX, a store-ready Android / iOS build.
9 · CUT LIST	NOT in this project: online leaderboard · shop, skins and coins · second playable hero · procedurally generated rooms · multiplayer / co-op · in-app purchases · story cutscenes · more than one boss.

10 · PLAN — WHAT IS READY EACH WEEK

W9	GDD, GitHub repo, mobile UI: menu, HUD with score, pause; SafeArea; Device Simulator on 3 devices
W10	Enemy AI: state machine (patrol → chase → attack), goblin archer, chasing slime
W11	Content + saving: 3 new levels, level select, PlayerPrefs / JSON save (levels, best score)
W12	Sound: Audio Mixer, music / SFX sliders in settings, new effects
W13	Visual effects: particles (hits, crystals, portal), hit flash, screen shake, 2D lights
W14–15	Team, ads, release build, store page — then the RK2 defense

ROLES	Solo until week 14 — I own every system: **programming** (player, enemies, boss, saving), **game design and levels** (text level maps, balance), **graphics and animation** (sprites, animator controllers), **sound and interface** (chiptune, menus, HUD), **release** (builds, README, video). In week 14 the project can join a team.
RISKS	Unity Remote on Windows lags → test the real UI in Device Simulator, build to a phone weekly when possible. Scope creep → the cut list above is final until week 14.